

CS362 Task 3: Kidney Matching Optimization Report

Overview

This project uses Google OR-Tools to solve the kidney exchange problem. The goal is to match as many patients with donors as possible through 2-way and 3-way swaps, taking into account blood types, match quality, and hard-to-match patients.

How It Works

The program finds all possible 2-way exchanges (A gives to B, B gives to A) and 3-way exchanges (A to B, B to C, C to A).

Each patient and donor pair can only be part of one exchange group.

The program gives points for each match based on three things:

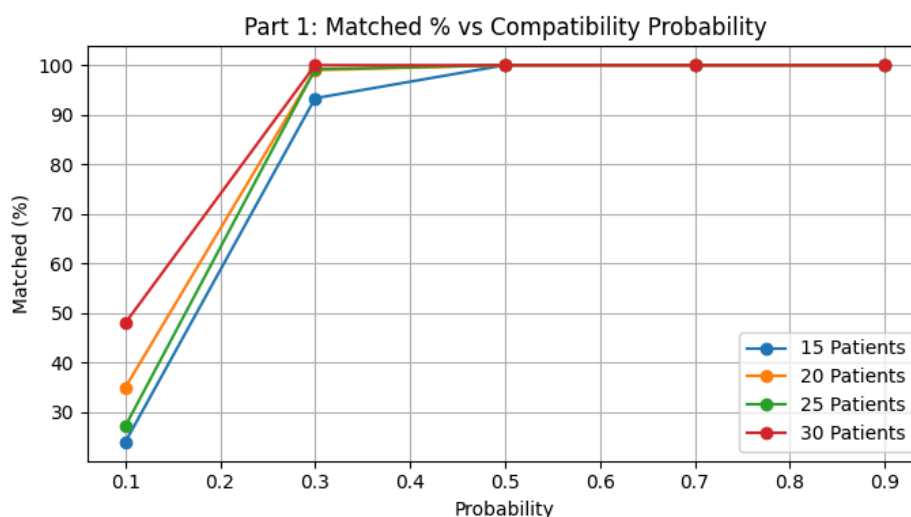
- A base score for making any match
- Extra points for high-quality matches
- Bonus points for helping hard-to-match (sensitized) patients

Results

Part 1: Basic Matching Test

What happened: When matches were rare (10% chance), larger groups of 30 patients matched twice as many people (48%) as smaller groups of 15 patients (24%).

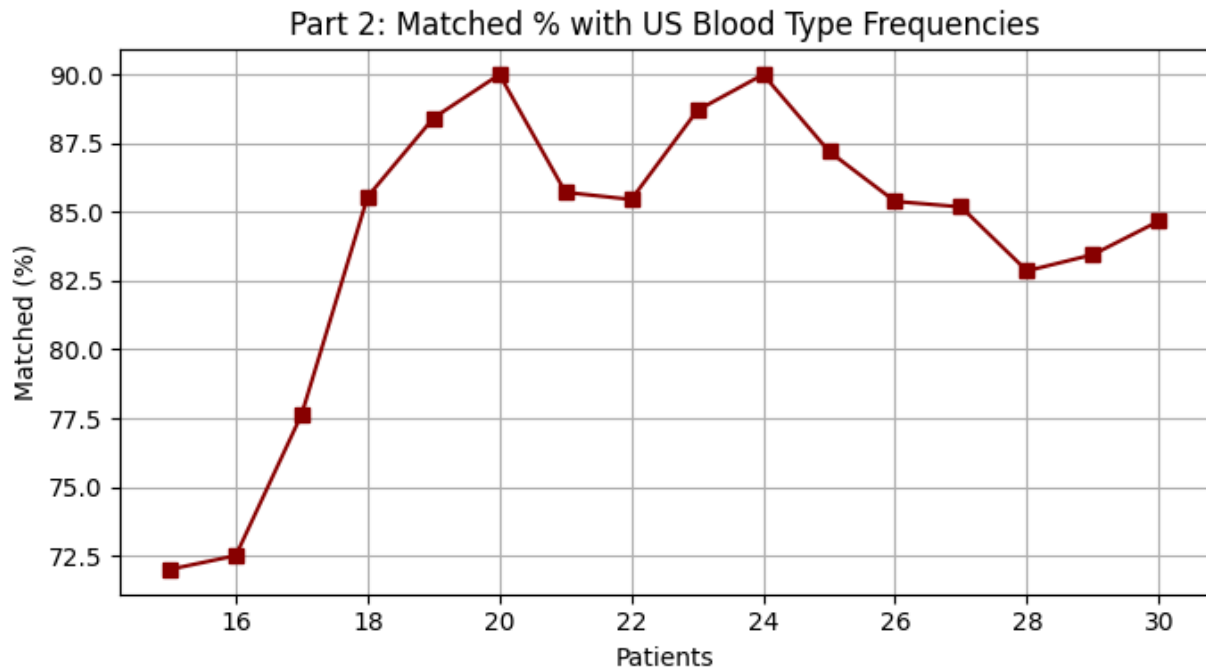
Main point: Having more people in the pool makes it much easier to find valid swaps. Once match chances reached 30% or higher, almost everyone got matched.



Part 2: US Blood Types Test

What happened: Using real US blood type numbers (O: 45%, A: 40%, B: 11%, AB: 4%) dropped match rates to between 83% and 90%.

Main point: Real life is harder than random math. Type O patients can only accept Type O kidneys, which creates a bottleneck that keeps some people from getting a match.



Part 3: Hard-to-Match Patients Test

What happened: Giving bonus points (from 0 to 10) for hard-to-match patients helped more of them get kidneys without lowering the total number of matches made overall.

Main point: The bonus successfully helps sensitive patients without hurting the rest of the group.

